ModelIt!

Teacher's Guide

The Water Crisis: Who Gets the Water?

When Rain Runs Out and People Keep Moving In

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-ESS2-2, 5-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS2-2, 5-ESS3-1

Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth. Obtain and combine information about ways individual communities use science ideas to protect the Earth's resources and environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how rainfall, population, and usage interact to determine water availability.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes** — Students model how water is distributed and how human activity affects its availability.
Crosscutting Concept	**Cause and Effect** — Students identify how population growth and reduced rainfall cause water crises.

CAST Testing Alignment

MULTIPLE CHOICE:

A town's population triples while rainfall stays the same. What is the most likely result?

CONSTRUCTED RESPONSE:

Explain how a drought and population growth together create a water crisis, using evidence from your model.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Fresh water makes up less than 3% of all water on Earth, and most of it is frozen in glaciers and ice caps. The tiny fraction that is accessible as groundwater and surface water must support all human communities, agriculture, and ecosystems.

Groundwater and Aquifers

Groundwater is water that has seeped through soil and rock into underground layers called aquifers. These aquifers are recharged by rainfall, but the refilling process takes years or decades.

Growing Demand

As populations grow, water demand increases for drinking, cooking, bathing, farming, and industry. Many aquifers are being pumped faster than rainfall can refill them.

Drought Amplifies the Problem

During drought, rainfall drops while demand stays the same or increases. This double pressure — less input, same output — can rapidly deplete groundwater reserves.

Real-World Water Crises

Cities like Cape Town, South Africa nearly ran out of water in 2018. Flint, Michigan had contaminated water. California regularly faces drought restrictions. Water crises are a global issue affecting billions.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Rainfall Amount, Population Size, Groundwater Level, Water Usage Rate, and Water Availability as the five system components.

E **Establish Relationships**
Students determine that Rainfall refills Groundwater (+), Population increases Usage (+), Usage depletes Groundwater (-), Groundwater supports Availability (+), and Population strains Availability (-).

V **Visualize & Model**
Students build the water crisis model showing competing inputs and outputs that determine whether a community has enough water.

E **Evaluate Outcomes**
Students run the Drought + Growth scenario and observe groundwater crashing and water availability dropping to crisis levels.

R **Revise & Extend**
Students add Desalination, Water Recycling, or Agricultural Irrigation to explore how technology and behavior can change outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What happens when the water runs out? Today we model a water crisis."*

Watch For

Show before/after photos of a real reservoir during drought.

Slide 2: Learning Objectives

2 min

Say: *"Today we explore how rainfall, population, and water usage interact."*

Watch For

Students read objectives and share what they know about droughts.

Slide 3: Big Question

2 min

Say: *"This is happening right now in communities around the world."*

Watch For

Quick discussion: Has anyone experienced water restrictions?

Slide 4: LEVER Framework

2 min

Say: *"Our model will have two inputs competing against each other."*

Watch For

Preview how rainfall adds water while population removes it.

Slide 5: Activity 1: Components

8 min

Say: *"Rainfall adds water to the system. Population takes it away. What happens in between?"*

Watch For

Students sort components and predict which are external vs internal.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice that Population has BOTH a positive and negative effect — it increases usage but decreases availability."*

Watch For

Guide students through all five relationships.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when rain drops and population rises? Let's watch the crisis unfold."*

Watch For

Students run Drought + Growth scenario and record observations.

Slide 8: Discoveries

5 min

Say: *"Water availability is a balance — when inputs fall below outputs, crisis follows."*

Watch For

Class discussion: What solutions exist?

Slide 9: STEM Challenge

10 min

Say: *"Design a plan to keep Dry Creek from running out of water."*

Watch For

Groups create plans, present to class for feedback.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Rainfall Amount, Population Size
- **INTERNAL:** Groundwater Level, Water Usage Rate, Water Availability

Relationships

- **Rainfall Amount to Groundwater Level:** POSITIVE (+) — More rainfall seeps into the ground and refills underground aquifers.
- **Population Size to Water Usage Rate:** POSITIVE (+) — More people means more water is used for drinking, bathing, and farming.
- **Water Usage Rate to Groundwater Level:** NEGATIVE (-) — Pumping water out of the ground faster depletes aquifer levels.
- **Groundwater Level to Water Availability:** POSITIVE (+) — Higher groundwater levels mean more water is available for the community.
- **Population Size to Water Availability:** NEGATIVE (-) — More people competing for the same water supply reduces availability per person.

Key Discoveries

- Groundwater refills slowly from rainfall but depletes quickly with high usage
- A growing population increases water demand even when rainfall stays the same
- Droughts are most dangerous in areas with large, growing populations
- Water availability depends on the BALANCE between rainfall input and usage output

Multiple Choice Post-Assessment Answer Key

1. In the water crisis model, what happens during the Drought + Growth scenario (Rainfall at 20%, Population at 80%)?

Correct: B) Groundwater Level drops rapidly because water usage far exceeds rainfall refill

Correct! With low rainfall, very little water is being added to the aquifer. But a large population keeps pulling water out at a high rate. Usage far exceeds refill, so groundwater drops rapidly and water availability crashes.

2. What type of relationship exists between Population Size and Water Availability?

Correct: C) Negative: more people competing for the same supply reduces water availability per person

That's right! When more people compete for the same water supply, each person gets less. Population growth strains the system unless rainfall also increases. This is a negative relationship.

3. Why does groundwater recovery take much longer than groundwater depletion?

Correct: B) Because rainfall seeps into aquifers slowly over time, while pumping can drain them quickly

Correct! Rainwater must slowly filter through soil and rock to reach underground aquifers, a process that takes years. But pumps can extract water much faster than this natural refill rate. That is why depletion is fast but recovery is slow.

4. A town's population triples while rainfall stays the same. Based on the model, what is the most likely result?

Correct: B) Water Usage Rate triples, groundwater depletes, and water availability drops sharply

Exactly! Population has a positive relationship with Water Usage Rate: more people use more water. But Rainfall has not changed, so the aquifer is losing water faster than it gains. Groundwater drops and water availability falls.

5. Which solution would the model predict is most effective for a town facing both drought and population growth?

Correct: C) Reducing water usage through conservation while exploring new supply sources like water recycling

Correct! The model shows that water availability depends on the balance between input and output. Conservation reduces demand (output), while water recycling increases effective supply (input). Attacking both sides of the balance is the most effective strategy.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"There is plenty of water because Earth is 70% water"	97% of Earth's water is saltwater. Only 3% is fresh, and most of that is frozen. Less than 1% is accessible groundwater and surface water.
"Rain always refills groundwater"	Rain refills aquifers, but the process takes years. If water is pumped out faster than rain can replace it, the aquifer depletes — just like withdrawing money faster than deposits.
"Water crises only happen in deserts"	Water crises can happen anywhere that demand exceeds supply. Even rainy regions can experience water shortages if population growth outpaces infrastructure.